

Coordinating Migration: Caring for Communities & Their Data

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ABSTRACT

This study inquires into the perspectives of the many groups who collect, manage, and use data generated as people migrate and settle in Canada. While their notions of care for communities and data may sometimes conflict, a range of stakeholders collaborate in their activities with data on immigration, including settlement service providers, migrant justice activists, immigration researchers, government staff and policymakers, and designers of digital systems that gather newcomers' data. As a connected yet distanced collective of stakeholders whose practices with data influence one another and the newcomers they study or serve, these same stakeholders also enact changes in their ways of using data and digital technologies in the context of experimentation with big data analytics, automation, and greater demands for data-based reporting and sharing. To this end, this research joins practical and theoretical discussions by working to strengthen webs of relations with greater capacity for care, informed reflection, and responsibility in the use of communities' data. Based on the lens of care, this project advances a critical approach to drastic shifts in information practices across areas of contemporary life, of which migration is a particularly pressing issue.

CCS CONCEPTS

• **Information systems**; • **Human-centered computing** → **Human computer interaction (HCI)**; • **Social and professional topics** → *User characteristics*;

KEYWORDS

care, ethics, mobilities, immigration, information practices, data

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1 MEDIATING MOBILITIES: INTRODUCTION

In this period of intensifying extremes, there are over 272 million international migrants in the world [16]. Climate change, protracted wars, discrimination or repression, dispossession from ancestral

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territories, and widening gaps between poverty and wealth are just some of the forces compelling people to move [38]. Yet, numbers alone are not sufficient to understand migrants' experiences. As a point of intersection between individuals, organizations, policies, technologies, and governments, *information practices* are sites where mobilities are mediated and understood.

To immigrate is to interact with numerous sociotechnical systems. Privacy rights have been eroded at and between borders, as surveillance measures to track movement are normalized [44]. We are seeing wide-ranging experimentation with information systems for border surveillance, policing, service provision, and the management of mobility. Consider the policy experiment to use genetic data collection and testing to vet "suspicious" asylum claims of applicants in the United Kingdom, amidst criticism from the scientific community [4]; data scientists' proposals to establish algorithmic mechanisms to assign refugees to resettlement locations in the United States and elsewhere [3]; and the implementation of automated drone surveillance systems used in European seas and skies that combine military and policing approaches, using risk-based taxonomies to assess and profile migrants [1]. In many instances, the uptake in these delegated and data-driven approaches goes along with national populism by states that seek to assess the "desirability" of would-be citizens. As part of these systems, people who cross borders are continually asked to provide proof of identity, submit to tracking to access government supports and services, and comply with pathways to permanent residence which are mediated through the use of digital data and sociotechnical systems. Despite concerns for the restriction and securitization of mobilities across borders, the many groups who cooperate in this context often share motivations to support migrants' wellbeing (e.g., humanitarian organizations, international aid agencies, policymakers, service providers, or scholars). Understanding these groups as networked moves this project to ask questions about the ways in which the work of caring for migrants and their data is distributed.

2 MIGRATION AS MATTER OF CARE: RELEVANCE

As individuals move through the migration process, countless digital traces are generated. Governments, institutions, and researchers are increasingly interested in collecting and analyzing these traces. The practices of generating, collecting, storing, analyzing, manipulating, sharing, and caring for these traces are important to study as these practices shape the possibilities for human mobility. However, to whom newcomers' data belongs and how it should be cared for by assemblages of people and technologies is less evident. My doctoral research investigates these questions of care for data in

dialogue with five communities who are involved with immigration and settlement in Canada.

Through this project, I suggest that it is not a problem in itself if multiple groups have varying understandings of care for communities, care for data, and information practices. Yet, from differences in approaches to data, there can be frictions, implications, and outcomes that are inequitably distributed and can contribute to potential harms or unintended consequences. This project explores different perspectives as part of a space for inquiry and communication, rather than a problem to be solved. If conversations within this space are not engaged, there are risks of overlooking the red lines of practice — that is, where boundaries and responsibilities need to be articulated and discussed between actors that are connected, but whose coordination is not necessarily explicit nor equitable.

3 INQUIRING INTO INFORMATION PRACTICES: RESEARCH QUESTIONS

This project attends to the multiplicity of perspectives on caring for communities and their data. The research questions that guide my project inquire into RQ1) the range of information practices enacted by different groups who collect and analyze newcomers' data, RQ2) if and how these groups see their practices shifting, RQ3) what they would like to learn from newcomers' data, RQ4) concerns for how newcomers' data is used, RQ5) how they suggest that newcomers' data should be cared for ethically, and RQ6) their sense of whether their practices are aligned with other groups who work with newcomers' data. To respond to these questions my research draws together the interests, concerns, and ideas of care articulated by experts from five interconnected groups: 1) settlement service providers, 2) migrant justice activists, 3) immigration researchers, 4) government staff and policymakers, and 5) designers of digital systems oriented towards supporting newcomers.

4 WEAVING WITH THE LITERATURE: RELATED WORK

As they negotiate migration, numerous stakeholders across scales — from individual, organizational, governmental, international, to planetary scales — need to interact with one another through their use of information and technologies. Yet, accelerating mobilities of people and data are not an inherent moral good, nor a given [38]. To examine the convergence of mobilities, power, and information practices, this study brings together theories of information practice with ethical orientations from feminist and Indigenous theories.

Theories of information practice direct attention to how people use tools to create, access, collect, manage, make sense of, and share information to meet their own and others' needs [2, 10, 13, 26, 31, 34, 43]. As a theoretical approach, this lens widens the frame on the many human agents as well as sociotechnical assemblages that co-participate in ways of using data, information, and knowledge, influencing one another in the process. Within these assemblages, actively shifting or transitioning information practices is known to be difficult [37, 45]. Enacting change in information practices is challenging even when groups of people are motivated by deeply held concerns and values as they push back on dominant or destructive processes, such as environmental extraction, overconsumption, and unsustainable lifeways [30, 31].

In information practice research, migrants or newcomers' interactions with information and tools have been a strong focus of research [e.g., 6, 10, 24].¹ Existing work richly examines the design and use of digital technologies, showing how these systems can meet information needs of migrants at various stages in their process of resettling [e.g., 25]. Complexities in how newcomers' agency is negotiated, enabled, and constrained by a range of interconnected actors, information, and technologies is an issue that will continue to develop [e.g., 10, 21, 36, 39].

There is strong evidence of the contested use of migrants' data at and between borders, whether for humanitarian aid, or other purposes such as immigration policy enforcement [32, 35]. This work suggests that rapid and unchecked experimentation with immigration data is already a norm across different countries. According to immigration, legal, and political science scholars, borders are also being externalized from and internalized within countries [e.g., 28, 29, 44]. Concerns with bodyworn cameras, biometrics, border sensors, and drones have been raised in these contexts [22, 44]. Pervasive digital bordering systems may co-opt citizens to become part of surveillance and policing apparatuses. Staff in public organizations and humanitarian agencies, for instance, are sometimes recruited into determining migrant's legitimacy [1, 19]. Bordering systems can be based on data gathered about migrants or newcomers, yet re-purposed by states or by private corporations contracted by states, raising significant human rights concerns [1].

What this previous work with migrants and other communities shows is that information practices, including the ethical values that intersect with practices, are a collaborative endeavour. In the context of data about immigration, ethical and practical responsibilities are distributed across organizations, types of expertise, sociotechnical systems, geographic locations, and distinct cultures of work. Deserving of further study are the ways newcomers' data can be created, collected, and processed with care by the many groups of people involved. This project therefore foregrounds care-taking relations, informed by Indigenous and feminist theories — specifically, ethics of care *and* caretaking relations developed in theories of Indigenous data governance, sovereignty, stewardship, and principles of caring for knowledge [e.g., 7–9, 12, 23, 27, 40, 42]. Lessons from scholars who have already studied caring for communities and their records can be returned to [e.g., 23]. Given these foundations in the literature, this project turns attention to how these ethical and theoretical traditions can invite reflections on how newcomers and their data are cared for.

5 MANY VOICES: METHODOLOGY

Through semi-structured interviews in the first phase of my project I have inquired into the interconnected information practices of settlement service providers, migrant justice activists, immigration researchers, government staff and policymakers, and designers of digital systems that gather newcomers' data.² In the second

¹The term migrant is often used broadly to be inclusive of immigrants, refugees, asylum seekers, and other groups of displaced peoples. The term newcomer is preferred in this project, and more specifically refers to individuals who have recently arrived in Canada regardless of their immigration status.

²The ethics of care orientation of my project informed my decision not to recruit a group of individuals currently navigating the process of migration. From my personal experiences and other research on migration, I understand that newcomers have considerable demands on their labour and time. However, lived experiences were

and final phase of my dissertation project, a narrative and visual resource is being co-designed with Olúwásọlá Kẹhíndé Olówó-Aké in the form of a card deck, in addition to other written outputs. The resource will represent insights generated through the project by offering questions about how to care for communities and their data, power relations, and points to pause in decision making, thereby aiming to support dialogues on how to act differently when caring with, about, and for this type of data.

6 INTERPRETATION & KNOWLEDGE EXCHANGE ACTIVITIES: EVALUATION PLAN

In the mode of exploratory research, this project seeks to widen understandings of involved stakeholders, and their concerns, interests, and practices. Stories and themes are interpreted in participants' contributions, following Braun and Clarke's [5] approach to thematic analysis and narrative analytic strategies from Frank [17]. The research is therefore assessed in relation to the foundations that it grows out of: It will be considered successful in the degree to which it investigates the terrain of stakeholders who are caretakers for newcomers' data, its written outputs extend existing theories of information practice, the methodology meets its ethical commitments, and its findings are deemed credible and ecologically valid to participants, assessed through member checking and knowledge exchange activities. During knowledge exchange workshops, I will invite feedback from participants on my interpretations in order to check findings and pilot the narrative and visual resource that supports dialogues on data practices.

7 LEARNINGS IN PROGRESS: PRELIMINARY RESULTS

Learnings from the project show signs of deepening understandings of shifting information practices and values negotiated across groups. Through interviews with 14 participants, I have had conversations about their understanding of their relationships with other groups who study and support immigration, their ideas of care for data, and their ways of employing information and technologies. The research has identified these groups as stakeholders implicated in collecting, managing, using, and actively changing practices of using newcomers' data, appreciated as a web of interconnected actors. Most participants identified the Immigration Contribution Agreement Management Environment (iCARE), a Canadian government database, as a system where many of their activities with immigration data are linked. The iCARE database was discussed by participants as a space where their concerns for data use are concentrated, and a challenging and changing digital system in urgent need of redesign. Some participants noted that iCARE's inequitable flows of data mean that the government amasses vast collections, linked across Canada and with potential for unknown future use. At the same time, individuals and communities who input data are limited in their ability to measure what matters to them through

still offered by many participants, who were once immigrants themselves and whose professional work is oriented towards representing and supporting immigrants on an everyday basis. It is my intention to engage more deeply with newcomers in future work when I am in a position to coordinate longer-term partnerships with immigrant-serving organizations.

iCARE, tied to service providers' ability to demonstrate outcomes of their work through quantitative evidence.

Early steps in analysis show signs that care for newcomers' data — with the iCARE database, and other digital and non-digital systems — are influenced by cultural factors, professional identities and norms, and the dynamics of material resourcing. Participants most proximal to serving newcomers, for example, pointed towards the demands to produce and work with data while having little access to relevant funding for staffing, systems, and technologies. Insights and stories from interviews also indicate that some stakeholders hold more decision-making power to select techniques and technological approaches than other stakeholders, and have greater capacity to decide what immigration data to collect and what to withhold from other groups.

8 JOINING CONVERSATIONS ABOUT CARING FOR COMMUNITIES & THEIR DATA: IMPLICATIONS

This project examines how information practices can be enacted that are not only functional, but also offer opportunities to care for current and future members of Canadian society. To this end, the research makes contributions by connecting theoretical concepts of care and ethical responsibilities with information practice scholarship [20, 42, 46]. This project also brings considerations of the consequences of applying algorithmic, autonomous, and intelligent technologies to human mobility [e.g., 14, 15, 18, 33]. Pragmatically, the narrative and visual card deck will centre the perspectives of stakeholders proximal to newcomers' experiences and their concerns. The card deck will also be a means of sharing preliminary findings back with participants, exchanging and building on knowledge generated through the project. Once designed, the card deck will invite cross-cultural, imaginative, and ethical reflections on how to care for newcomer communities and their data.

9 CONCLUSION

CSCW scholarship has drawn attention to the performance of caring labour with sociotechnical systems and the data generated through these activities [11, 41]. Coordination, collaboration, co-operation, and care in stewarding the data of others are matters of concern that will only grow. It is time to bring more attention to sites where caretaking information practices are enacted across scales and stakeholder groups, addressing the most pressing issues we face, including displacement and migration. To close with the words of migrant justice scholar Harsha Walia, "We can weave solidarities through the lens of abundance, rather than scarcity, and celebrate the interdependence of the particularities of our humanities. We can reconstruct egalitarian social ecosystems, where we cultivate depropertied, decommodified, decarceral, decolonial, and democratic relations" [44, p. 215]. In these border-transcending perspectives we may consider how different ideas of kin making, caretaking, home making, and world making are possible through our information practices.

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REFERENCES

- [1] E Tendayi Achiume. 2020. Racial discrimination and Emerging Digital Technologies: A Human Rights Analysis. <https://undocs.org/en/A/HRC/44/57>.
- [2] Theresa Dirndorfer Anderson. 2007. Settings, arenas and boundary objects: Socio-material framings of information practices. *Information Research* 12, 4 (2007).
- [3] Kirk Bansak, Jeremy Ferwerda, Jens Hainmueller, Andrea Dillon, Dominik Hangartner, Duncan Lawrence, and Jeremy Weinstein. 2018. Improving refugee integration through data-driven algorithmic assignment. *Science* 359, 6373 (2018), 325–329.
- [4] Ruha Benjamin. 2016. Informed refusal: Toward a justice-based bioethics. *Science, Technology, & Human Values* 41, 6 (2016), 967–990.
- [5] Virginia Braun and Victoria Clarke. 2006. Using Thematic Analysis in Psychology. *Qualitative Research in Psychology* 3, 2 (2006), 77–101.
- [6] Nadia Caidi, Danielle Allard, and Lisa Quirke. 2010. Information practices of immigrants. *Annual review of information science and technology* 44, 1 (2010), 491–531.
- [7] Stephanie Russo Carroll, Ibrahim Garba, Oscar L Figueroa-Rodriguez, Jarita Holbrook, Raymond Lovett, Simeon Materechera, Mark Parsons, Kay Raseroka, Desi Rodriguez-Lonebear, Robyn Rowe, et al. 2020. The CARE Principles for Indigenous Data Governance. *Data Science Journal* 19, 1 (2020).
- [8] Sasha Costanza-Chock. 2020. *Design justice: Community-led practices to build the worlds we need*. MIT Press, Cambridge, MA.
- [9] Maria Puig de la Bellacasa. 2017. *Matters of care: Speculative ethics in more than human worlds*. University of Minnesota Press, Minneapolis, MN.
- [10] Antonio Diaz Andrade and Bill Doolin. 2019. Temporal enactment of resettled refugees' ICT-mediated information practices. *Information Systems Journal* 29, 1 (2019), 145–174.
- [11] Catherine D'Ignazio, Erhardt Graeff, Christina N. Harrington, and Daniela K. Rosner. 2020. Toward Equitable Participatory Design: Data Feminism for CSCW amidst Multiple Pandemics. In *Conference Companion Publication of the 2020 on Computer Supported Cooperative Work and Social Computing*. ACM, New York, NY, 437–445.
- [12] Catherine D'Ignazio and Lauren F Klein. 2020. *Data feminism*. MIT Press, Cambridge, MA.
- [13] Paul Dourish and Ken Anderson. 2006. Collective information practice: Exploring privacy and security as social and cultural phenomena. *Human-computer interaction* 21, 3 (2006), 319–342.
- [14] Paul Dourish and Edgar Gomez Cruz. 2018. Datafication and data fiction: Narrating data and narrating with data. *Big Data & Society* 5, 2 (2018), 1–10.
- [15] Virginia Eubanks. 2018. *Automating inequality: How high-tech tools profile, police, and punish the poor*. St. Martin's Press, New York, NY.
- [16] International Organization for Migration. 2020. World migration report 2020. www.iom.int/wmr.
- [17] Arthur W Frank. 2010. *Letting stories breathe: A socio-narratology*. University of Chicago Press, Chicago, IL.
- [18] Anna Lauren Hoffmann. 2018. Making data valuable: Political, economic, and conceptual bases of big data. *Philosophy & Technology* 31, 2 (2018), 209–212.
- [19] Lotta Hultin and Lucas Introna. 2019. On receiving asylum seekers: Identity working as a process of material-discursive interpellation. *Organization Studies* 40, 9 (2019), 1361–1386.
- [20] Nassim JafariNaimi, Lisa Nathan, and Ian Hargraves. 2015. Values as hypotheses: design, inquiry, and the service of values. *Design issues* 31, 4 (2015), 91–104.
- [21] Dragana Kaurin. 2019. Data protection and digital agency for refugees. (2019).
- [22] Mark Latonero, Keith Hiatt, Antonella Napolitano, Giulia Clericetti, and Melanie Penagos. 2019. Digital identity in the migration & refugee context. *Data & Society* (2019).
- [23] Kimberley L Lawson. 2004. *Precious fragments: First Nations materials in archives, libraries and museums*. Master's thesis. University of British Columbia, Vancouver, Canada.
- [24] Jessa Lingel. 2015. Information practices of urban newcomers: An analysis of habits and wandering. *Journal of the Association for Information Science and Technology* 66, 6 (2015), 1239–1251.
- [25] Carleen Maitland. 2018. *Digital lifeline?: ICTs for refugees and displaced persons*. MIT Press, Cambridge, MA.
- [26] Pamela J McKenzie. 2003. A model of information practices in accounts of everyday-life information seeking. *Journal of Documentation* 59, 1 (2003), 19–40.
- [27] Annemarie Mol, Ingunn Moser, and Jeannette Pols. 2010. *Care in practice: On tinkering in clinics, homes and farms*. transcript Verlag, Bielefeld, Germany.
- [28] Petra Molnar. 2020. Technological Testing Grounds: Migration Management Experiments and Reflections from the Ground Up. <https://edri.org/wp-content/uploads/2020/11/Technological-Testing-Grounds.pdf>.
- [29] Petra Molnar and Lex Gill. 2018. Bots at the gate: a human rights analysis of automated decision-making in Canada's immigration and refugee system. (2018).
- [30] Lisa P Nathan. 2009. *Ecovillages, sustainability, and information tools: An ethnography of values, adaptation, and tension*. Ph.D. Dissertation. Seattle, WA.
- [31] Lisa P Nathan. 2012. Sustainable information practice: An ethnographic investigation. *Journal of the American Society for Information Science and Technology* 63, 11 (2012), 2254–2268.
- [32] Bryce Clayton Newell, Sara Vannini, and Ricardo Gomez. 2020. The information practices and politics of migrant-aid work in the US-Mexico borderlands. *The Information Society* 36, 4 (2020), 199–213.
- [33] Safiya Umoja Noble. 2018. *Algorithms of oppression: How search engines reinforce racism*. NYU Press, New York, NY.
- [34] Michael Olsson and Annemaree Lloyd. 2017. Being in place: embodied information practices. *Information Research* 22, 1 (2017).
- [35] Annalisa Pelizza. 2020. Processing alterity, enacting Europe: Migrant registration and identification as co-construction of individuals and politics. *Science, Technology, & Human Values* 45, 2 (2020), 262–288.
- [36] Dina Sabie and Syed Ishtiaque Ahmed. 2019. Moving into a technology land: exploring the challenges for the refugees in Canada in accessing its computerized infrastructures. In *Proceedings of the 2nd ACM SIGCAS Conference on Computing and Sustainable Societies*. 218–233.
- [37] Bryan Semaan, Lauren M Britton, and Bryan Dosono. 2017. Military masculinity and the travails of transitioning: Disclosure in social media. In *Proceedings of the 2017 ACM Conference on computer supported cooperative work and social computing*. 387–403.
- [38] Mimi Sheller. 2018. *Mobility justice: The politics of movement in an age of extremes*. Verso, London, United Kingdom.
- [39] Reem Talhouk, Lizzie Coles-Kemp, Rikke Bjerg Jensen, Madeline Balaam, Andrew Garbett, Hala Ghattas, Vera Araujo-Soares, Balsam Ahmad, and Kyle Montague. 2020. Food Aid Technology: The Experience of a Syrian Refugee Community in Coping with Food Insecurity. *Proceedings of the ACM on Human-Computer Interaction* 4, CSCW2 (2020), 1–25.
- [40] Kim TallBear. 2019. Caretaking relations, not American dreaming. *Kalfou* 6, 1 (2019), 24–41.
- [41] Austin Toombs, Laura Devendorf, Patrick Shih, Elizabeth Kaziunas, David Nemer, Helena Mentis, and Laura Forlano. 2018. Sociotechnical Systems of Care. In *Companion of the 2018 ACM Conference on Computer Supported Cooperative Work and Social Computing*. ACM, New York, NY, 479–485.
- [42] Joan C Tronto. 1998. An ethic of care. *Generations: Journal of the American Society on Aging* 22, 3 (1998), 15–20.
- [43] Kimmo Tuominen, Sanna Talja, and Reijo Savolainen. 2005. The social constructionist viewpoint on information practices. In *Theories of information behavior*. Information Today, 328–333.
- [44] Harsha Walia. 2021. *Border and Rule: Global migration, capitalism, and the rise of racist nationalism*. Haymarket Books, Chicago, IL.
- [45] Rebekah Willson. 2019. Transitions Theory and Liminality in Information Behaviour Research: Applying New Theories to Examine the Transition to Early Career Academic. *Journal of Documentation* 75, 4 (2019), 838–856.
- [46] Iris Marion Young. 2010. *Responsibility for justice*. Oxford University Press, Oxford, United Kingdom.